Manuscripts considered — osteosarcoma-mets-dll3-h7r2

Master inventory: every paper reviewed in this case

Generated 2026-05-05

Manuscripts considered — osteosarcoma-mets-dll3-h7r2

Master inventory: 3 clinical (2 included, 1 considered & excluded) + 5 pre-clinical (4 included, 1 considered & excluded) + 1 additional trial publications/registrations. One row per manuscript, sorted by year (newest first). Excluded rows are papers a Libby agent reviewed and chose NOT to feed to the board, with the exclusion reason captured.

Report		Type	Inclusion		Indication / model	Design	n	Effect size	Variance	Toxicities (type, n/N, rate)	Case fit	Notes
(2025) — N Engl J Med	PMID 40454646	clinical	included	tarlatamab (DLL3 × CD3 BiTE)	SCLC after platinum-based chemo (2L); or SCLC; ECOG 0-1	randomized phase 3	509	0.6 HR — OS (HR vs chemo)	95% CI 0.47–0.77; 95% CI; median OS 13.6 vs 8.3 mo; p=<0.001	CRS (n=254, 56%); CRS G>=3 (n=254, 1%); ICANS-like neurologic event (n=254, 12%); treatment-related AE G>=3 (n=254, 24%); neutropenia G>=3 (n=254, 3%)		Confirmatory phase-3 OS benefit in SCLC. Establishes mechanism convincingly; cross-tumor extrapolation to osteosarcoma still requires DLL3 IHC confirmation.
—/— (2025) — record / ASCO 2025 abstract TPS2700	—	trial	included	tarlatamab (DLL3 × CD3 BiTE)	advanced solid tumors (basket) (2L+); DLL3 IHC ≥25% (stage 1) or ≥1% (stage 2)	single-arm phase 2 basket (Simon two-stage)	29	— — ORR at 18 months (RECIST 1.1)	—	—	partial	(registration only — no peer-reviewed publication yet; toxicity table not extracted)
(2023) — N Engl J Med	PMID 37861218	clinical	included	tarlatamab (DLL3 × CD3 BiTE)	previously treated SCLC (2L+); SCLC after platinum-based therapy and ≥1 prior line; ECOG 0-1; brain mets allowed if treated	open-label phase 2	220	40 % — ORR (BICR)	95% CI 29–52; 95% CI in 10mg cohort	CRS (n=100, 51%); CRS G>=3 (n=100, 1%); ICANS-like neurologic event (n=100, 10%); treatment-related AE G>=3 (n=100, 30%); treatment discontinuation due to AE (n=100, 3%)		Pivotal trial driving FDA accelerated approval. Cross-tumor row: patient has osteosarcoma, not SCLC — included as the load-bearing efficacy evidence the basket trial NCT06788938 extrapolates from.

Zhang/Shi (2023) — Aging (Albany NY)	PMID 37179118	included	DLL3 pan-cancer expression atlas	TCGA pan-cancer transcriptomic + protein-atlas analysis	of DLL3 expression across tumor types and correlations with TMB, MSI, and tumor	33	moderate — DLL3 expression detectable across 18 cancer types; co-varies with immune infiltration in subsets. Sarcoma signal present but not	vs GTEx normal-tissue baseline; translatability: low	n/a (preclinical)	weak	Bulk RNA-level analysis. Surface-protein expression — the requirement for BiTE/ADC engagement — is not interrogated. Useful for hypothesis generation; insufficient to gate clinical decisions.
Yao/Pavel (2022) — Oncologist	PMID 35983951	included	DLL3 as therapeutic target — review	Literature review	overview of DLL3 biology, IHC and landscape (BiTEs, ADCs, CAR-T,	—	moderate — DLL3 is a antigen with low normal-tissue expression — attractive when present at adequate density. detection (e.g. SP347 IHC) is the consensus gating biomarker for DLL3-directed therapies.	vs —; translatability: med	n/a (preclinical)	weak	Focuses on neuroendocrine neoplasms. Does not establish osteosarcoma-specific evidence. Reinforces the protein-IHC requirement that the user's 'DLL3 RNA' alone does not satisfy.

(2021) — JTO	PMID 33508456	clinical	excluded — Drug by sponsor (AbbVie withdrew Rova-T after the TAHOE phase-3 SCLC trial showed worse OS vs topotecan). Mechanism ADC) is informative for the target rationale but the molecule itself is not procurable.	tesirine (Rova-T, DLL3 ADC)	—	—	—	—	—	—	—	Excluded: Drug discontinued by sponsor (AbbVie withdrew Rova-T after the TAHOE phase-3 SCLC trial showed worse OS vs topotecan). Mechanism (DLL3-targeted PBD-warhead ADC) is informative for the target rationale but the molecule itself is not procurable.

Hipp/Wolf (2020) — Clin Cancer Res	PMID 32694156	excluded — Original BiTE construct (AMG 757 / tarlatamab) in SCLC models — superseded by Giffin 2021 which includes the cytolysis data directly relevant to the IHC ≥1% / ≥25% threshold this case turns on.	tarlatamab (DLL3 × CD3 BiTE)	—	—	—	—	—	n/a (preclinical)	—	Excluded: Original BiTE construct (AMG 757 / tarlatamab) characterization in SCLC models — superseded by Giffin 2021 (pmid:35983951) which includes the cytolysis data directly relevant to the IHC ≥1% / ≥25% threshold this case turns on.
(2019) — Clin Cancer Res program)	—	included	tarlatamab (DLL3 × CD3 BiTE)	DLL3-positive SCLC xenograft + humanized PBMC co-culture	Bispecific antibody co-engages DLL3 on tumor cells and CD3 on T cells, redirecting cytotoxic T-cell killing.	n=8 in n=3	strong — cytolysis demonstrated; ≥1% surface DLL3 by IHC sufficient for activity in SCLC models. Tumor regression sustained beyond treatment cessation in xenograft.	vs vehicle + PBMCs without bispecific; translatability: med	n/a (preclinical)		All published mechanistic data are SCLC-derived. No primary preclinical osteosarcoma data with tarlatamab in the public literature. Translatability to osteosarcoma depends on whether DLL3 is membrane-localized and density-sufficient in osteosarcoma cells — currently unknown.

Wang/Zhu (2018) — J Exp Clin Cancer Res	PMID 29475441	included	pathway in (target rationale)	Osteosarcoma cell lines (MG-63, U2OS, Saos-2); primary osteosarcoma samples	axis implicated in stemness and SOX2OT lncRNA upregulates signaling.	n=3 per	moderate — EGCG + doxorubicin combination suppresses Notch3/DLL3 signaling and reduces stemness in suggests pathway is operative, not just expressed.	vs vehicle + scrambled siRNA / parental cell line; translatability: low	n/a (preclinical)	weak	Pathway-level mechanistic study; does NOT measure cell-surface DLL3 protein expression by IHC. Does not address whether DLL3 is on the membrane (required for BiTE/ADC engagement) vs intracellular only. The user's 'DLL3 RNA expression' input matches the type of evidence here — but this work cannot bridge the gap that DLL3-targeted therapies need.